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100%

QUIZ 3

TRANSPORT PHENOMENA II

CLOSED BOOK, TIME ALLOWED: 40 MIN

Consider a porous cylindrical catalyst with radius $r=R$ and length $z=L$. Species A diffuses into the catalyst. The zeroth-order reaction rate coefficient is k_0'' (moles of A generated per unit time per unit area is $-k_0''$). At the catalyst surface, concentration of A is specified as $c_A(R) = c_{A0}$. Assume:

- Stagnant system (neglect convection),
- Diffusion only happens in the radial direction,
- Steady state & isothermal conditions.

$$\frac{\partial}{\partial t} = 0 \quad \Delta T = 0$$

- ✓ 1- Given that the mole fraction of A in the catalyst is very small ($x_A \ll 1$), simplify the combined mole flux

$$\langle N_A \rangle = c_A \langle v_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$$

to

$$\langle N_A \rangle = -D_A \langle \nabla c_A \rangle$$

Recall: $\langle J_A^* \rangle = -c D_A \langle \nabla x_A \rangle$ where c is the total concentration.

- ✓ 2- Set up a shell balance to derive an ODE for the steady-state concentration profile of A as a function of r

$$\int_V dV \frac{\partial c_A}{\partial t} = - \int_S dS \langle N_A \rangle n + \int_V dV R_A$$

- ✓ 3- List 2 boundary conditions.

- ✓ 4- Solve the ODE and determine the integration constants.

Your answers may be in terms of the following constants:

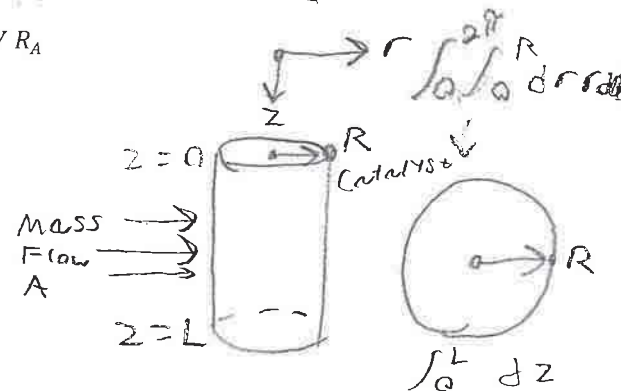
R and L

k_0'' : surface reaction rate constant in moles per unit time per unit area

a : available surface area per unit volume of the catalyst.

D_A : effective diffusivity of species A in the porous medium, where Fick's law in terms of effective diffusivity is

c_{A0} : molar concentration of A at the surface of the catalyst.



$$\begin{aligned}
 1) \langle N_A \rangle &= c_A \langle v_A \rangle = c_A \langle v_A^* \rangle + \langle J_A^* \rangle \\
 &= c_A \langle v_A^* \rangle - c D_A \langle \nabla x_A \rangle \\
 &\xrightarrow{x_A \ll 1} = \cancel{x_A c} \langle v_A^* \rangle - c D_A \langle \nabla x_A \rangle \\
 &= -c D_A \langle \nabla x_A \rangle \\
 &= -D_A \langle \nabla c_A \rangle
 \end{aligned}$$

$$2) \int_V dV \frac{\partial C_A}{\partial t} = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

$$0 = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

$$0 = 2\pi r L \langle N_A(r) | - 2\pi r L \langle N_A(r+\Delta r) | = 2\pi r L \langle k_0'' \rangle$$

$$0 = \frac{r N_A(r) - r N_A(r+\Delta r)}{r \Delta r} = k_0''$$

$$0 = - \frac{\partial(r N_A)}{r \partial r} = k_0'' \rightarrow \frac{\partial(r N_A)}{r \partial r} = - k_0''$$

$$* \nabla C_A = \frac{\partial C_A}{\partial r} \text{ only} \quad -k_0'' = \frac{r(-D_A \nabla C_A)}{r \partial r} = - \frac{1}{r} D_A \frac{\partial^2 (C_A r)}{\partial r^2}$$

$$\frac{\partial^2 (C_A r)}{\partial r^2} = \frac{k_0'' r}{D_A} \rightarrow \frac{\partial^2 (C_A r)}{\partial r^2} = \frac{k_0'' r}{2 D_A} + \frac{C_1}{r}$$

$$C_A = \frac{k_0'' r^2}{4 D_A} + C_1 \ln(r) + C_2 \checkmark$$

$$3) BC: C_A(R) = C_{A0} \quad (1) \checkmark$$

$$N_A(r=0) = \text{finite} \quad (2) \checkmark$$

$$4) (1) C_{A0} = \frac{k_0'' R^2}{4 D_A} + C_1 \ln(R) + C_2$$

$$(2) 0 = 0 + \frac{C_1}{0} \rightarrow C_1 = 0 \checkmark$$

$$C_{A0} = \frac{k_0'' R^2}{4 D_A} + C_2 \rightarrow C_2 = \frac{C_{A0}(4 D_A)}{k_0'' R^2} \checkmark$$

$$C_A = \frac{k_0'' r^2}{4 D_A} + \frac{C_{A0}(4 D_A)}{k_0'' R^2}$$